

Thermodynamics

Grau en Enginyeria Matemàtica i Física

Joan Rosell Llompart

Departament d'Enginyeria Química
Universitat Rovira i Virgili, Tarragona

joan.rosell@urv.cat

Second Quatrimester 2023-24

1. Basic definitions.

What does Equilibrium Thermodynamics study?

The properties of so called *N-body* systems can be divided into:

- ▶ **Microscopic** properties: atoms and molecules, orbitals and electrons, Quantum Mechanics, etc.
- ▶ **Macroscopic** properties: collective properties, resulting from the behaviour of many particles, pressure and temperature, etc.

Thermodynamics is the science which establishes relationships between macroscopic properties of a system.

Thermodynamics is a science of states, not of processes, despite being called Thermo-DYNAMICS.

Thermodynamic systems must contain a large number of particles (atoms, molecules, ...) for it to be applicable.

What does Thermodynamics study?

When averaging the atomic coordinates, only a few coordinates survive the averaging that Mechanics is interested in. For example in a solid: its mass, its center of mass, its moment of inertia, the deformation components, etc.

Mechanics is the study of the behavior of such surviving coordinates. It happens similarly with **electricity**.

Thermodynamics is the study of the macroscopic consequences of the countless atomic coordinates which, by virtue of the statistical averaging, do not appear explicitly in the macroscopic description of the system. Callen.

For example, we can transfer energy to a solid by deforming it, via **mechanical work**. Energy is thus transferred to the **macroscopically observable modes**. On the other hand, energy can also be transferred to the **hidden atomic modes**, via **heat**.

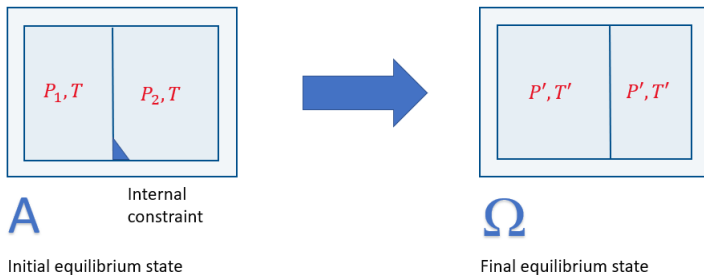
The fundamental problem of Thermodynamics

The specific problems of physics (or engineering) that can be solved by thermodynamics fall into three types:

1. **For a given process, with prescribed internal constraints and certain boundary conditions (walls), when we free it from these internal constraints, what will be the final equilibrium state?**
2. To cause a certain change in the thermodynamic properties of a system, what external actions must be imposed?
3. Among all the possible processes that cause a certain change in the thermodynamic properties of a system, what are the efficiencies of each of them in relation to the sources at our disposal?

The fundamental problem of Thermodynamics

Fundamental problem: when we liberate the system from its internal constraint, starting from equilibrium state A, what will be the final equilibrium state Ω ?



1.1. Basic concepts and definitions

The system and its surroundings

- ▶ The **system**:
That part of space which we identify to be studied.
- ▶ The **boundaries (or walls)**:
The surfaces (physical or not) which bound the system.
- ▶ The **surrounding**:
The space outside of the system with which it can exchange physical and chemical properties.

1.1. Basic concepts and definitions

State variables

They are macroscopic variables of the system in an equilibrium state, which characterize it, and which we can determine (directly or indirectly).

They are not properties of this type the speed of a specific atom or its quantum state.

Some examples of the first ones are

- ▶ Thermodynamic temperature T
- ▶ Pressure P
- ▶ Volume V
- ▶ Internal energy U
- ▶ Entropy S
- ▶ ...

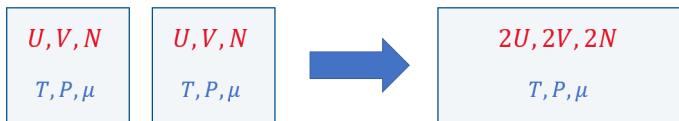
1.1. Basic concepts and definitions

Intensive and extensive variables

State variables are classified into two types:

- ▶ **Extensive:** Those which depend on the size of the system. Some examples are $U, S, V, N, H, \dots$
- ▶ **Intensive:** Those which do not depend on the size of the system. Some examples are $T, P, \mu, \dots$

Extensive and intensive variables



1.1. Basic concepts and definitions

Types of walls

Walls are classified according to the type of property they allow to pass through.

- ▶ **Diathermal/Adiabatic:** They allow/don't allow **heat** to pass through.
- ▶ **Mobile/Rigid:** They allow/don't allow changes in **volume**.
- ▶ **Permeable/Impermeable/Semipermeable:** They allow/don't allow the transfer of **matter**, or the transfer of some substances but not others.

1.1. Basic concepts and definitions

Types of walls

Additional nomenclature

- ▶ Systems bounded by diathermal and impermeable walls are called **closed** systems.
- ▶ Systems bounded by diathermal and (semi)permeable walls are called **open** systems.
- ▶ Systems bounded by adiabatic, rigid and impermeable walls are called **isolated** systems.

1.1. Basic concepts and definitions

Processes

The change from one (equilibrium) state to another is called **process** and it takes place in time.

Processes are initiated by a change in

- ▶ the **internal constraints** of the system, or
- ▶ in the **external conditions** (e.g., the temperature of a bath) which have an influence on the system, if its walls allow it (e.g., diathermal walls), or
- ▶ the **type of wall** (e.g., from adiabatic to diathermal).

1.1. Basic concepts and definitions

Processes

We distinguish processes according to two criteria:

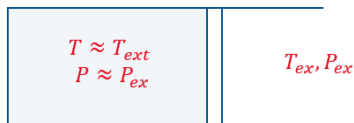
- ▶ With respect to their **temporal evolution**:
 - ▶ **Quasistatic process**: One which evolves infinitely slowly, in such a way that the system is always in its equilibrium state compatible with the conditions of the process.
 - ▶ **Non-quasistatic processes**: all the rest.
- ▶ With respect to their **reversibility**
 - ▶ The process for which an infinitesimal change in the external conditions can invert its direction is called **reversible**.
 - ▶ The process which requires a finite change in the external conditions is called **irreversible**.

1.1. Basic concepts and definitions

Processes

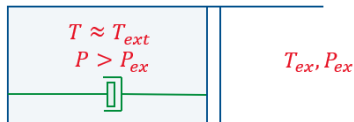
All reversible processes are quasistatic.

Commonly, the term *quasistatic* is used as synonym of *reversible*; however, strictly speaking, this is not correct. Let us see an example:



We vary the external pressure at an infinitely slow rate.

Quasistatic and REVERSIBLE



We vary the external pressure abruptly. The system evolves very slowly because it has a damper which slows down the piston.

Quasistatic and IRREVERSIBLE

1.2. Equilibrium states and existence of temperature

Postulate I

*There exist particular states called **equilibrium states** which, from a macroscopic point-of-view, are completely characterized by their internal energy U , their volume V and their composition, expressed by the number of moles of each of their components $N_1, N_2, \dots, N_n$*

Such states are characterized by the constancy of its variables in time, when neither the external conditions nor the internal constraints are modified.

Recall: Equilibrium states are perturbed by the change in a) constraints inside the system, b) the kind of wall (e.g., adiabatic to diathermal), or c) by the modification of the external conditions (in the surroundings with which the system interacts).

1.2. Equilibrium states and existence of temperature

Equilibrium states

Operational definition of temperature: Zeroth principle

Traditionally, this postulate has been considered *the zeroth principle* of Thermodynamics, which has been stated as follows:

If among three bodies, A, B i C, bodies A i B are separately in thermal equilibrium with C, then, A i B are also in thermal equilibrium with one another.

The property shared between A, B i C in thermal equilibrium is **temperature**.

A thermometer is, therefore, a system which, when placed in contact with a system A, when the two attain thermal equilibrium they share the same temperature. Then, we can indirectly measure such temperature by measuring the changes that are produced in one property of the thermometer: volume of mercury, or the pressure of a gas, etc.

1.2. Equilibrium states and existence of temperature

Temperature

Operationally, we have defined temperature as the *property shared by systems which are in thermal contact, after reaching equilibrium*. Nonetheless, since Boltzmann and Maxwell, we have an interpretation of temperature in connection with the kinetic energy of the particles.

Kinetic theory

The distribution (pdf) of velocities of a gas in equilibrium at a temperature T looks like:

$$f(v_x) dv_x = \left(\frac{m}{2\pi k_B T} \right)^{1/2} e^{-mv_x^2/(2k_B T)} dv_x$$

where $k_B \equiv R/N_A$. This distribution is identical for the v_y i v_z components. Given that $\langle v_x^2 \rangle = k_B T/m$, the total kinetic energy of the atoms in the gas can be estimated as

$$U = \sum_{i=1}^{\mathcal{N}} \frac{1}{2} m v_i^2 = \frac{3}{2} \mathcal{N} k_B T = \frac{3}{2} NRT \quad \text{monoatomic gas (e.g., argon)}$$

This allows us to interpret the internal energy of a system as **energy stored in the system's degrees of freedom which cannot be observed macroscopically**.

1.2. Equilibrium states and existence of temperature

Temperature

There are two groups of temperature scales:

1. Celsius and Kelvin
2. Fahrenheit and Rankine

Celsius and Kelvin

The Celsius scale assigns 0°C to the temperature at which water melts, and 100°C to the temperature at which it boils. Since the zero value is placed at the melting temperature of water, positive and negative values are possible.

The Kelvin temperature scale was introduced to take advantage of the existence of a smaller temperature below which it is impossible to go. It is called *the thermodynamic temperature scale*.

The conversion between scales is as follows:

$$T(K) = T(^{\circ}\text{C}) + 273.15$$

1.2. Equilibrium states and existence of temperature

Temperature

Fahrenheit i Rankine

The Fahrenheit scale *aproximately* assigns the $0^{\circ}F$ to the melting temperature of brine (cat. salmorra) and $100^{\circ}F$ to the rectal temperature of a cow. To learn more, see [3].

With respect to the Celsius scale, the Fahrenheit scale has smaller units, while the zero is in a different place. The transformation between these scales is as follows:

$$T(^{\circ}F) = \frac{9}{5} T(^{\circ}C) + 32$$

The Rankine temperature scales was also introduced as a thermodynamic temperature scale, but with finer degress than the Fahrenheit scale.

The relationship between Rankine and Kelvin is as follows:

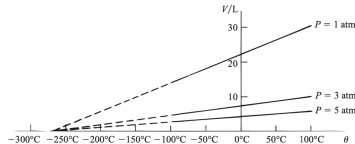
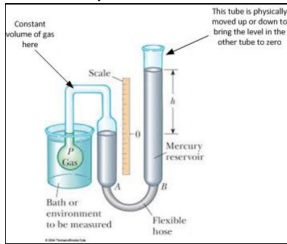
$$T(R) = \frac{9}{5} T(K)$$

1.2. Equilibrium states and existence of temperature

Existence of absolute zero

Consider a gas thermometer. In this, we take a fixed volume of gas and measure the change in pressure as the temperature changes. Pressure is therefore an indirect measure of temperature and, for this reason, we will have to calibrate the thermometer. Temperature measurements with different gases suggest that the pressure cancels out at a temperature $T = -273.15^\circ\text{C}$. Below this temperature there is no thermal agitation and the system would have no internal energy. That is why this minimum temperature cannot be exceeded, and the thermodynamic scales take it

as zero.



1.3. The mol

Massa atòmica

The atomic mass unit **amu** is defined as one-twelfth the mass of a carbon 12 atom (12 being the species' mass number). This makes it roughly the mass of a neutron.

$$1 \text{ amu} = \frac{1}{12} m(^{12}\text{C})$$

The atomic mass of water will therefore be

$$\frac{m(\text{H}_2\text{O})}{\frac{1}{12} m(^{12}\text{C})} = 18.015 \text{ amu}$$

The decimals arise due to the population of atomic isotopes (same charge but different mass). For hydrogen they are protium, deuterium, and tritium.

The **Dalton** is the unit of atomic mass used in biochemistry, and corresponds to 1 amu.

1.3. The mol

Avogadro's number

Avogadro introduced the concept of the number of particles based on the hypothesis that the pressure of an ideal gas in a container had to be proportional to the number of particles it contained.

From this, the *mol* or *gmol* (*gram-mol*) was defined.

A gmol is the number of atoms of ^{12}C that are in 12 g of this species.

Such number (given approximately next) is known as *Avogadro's number*

$$N_A = 6.02 \times 10^{23}$$

The following are also used:

1. **kgmol**: the number of ^{12}C atoms in 12 kg. Equivalent to 1000 mol.
2. **lbmol**: The number of ^{12}C atoms in 12 lb. Equivalent to 453.59237 mol.

A mol contains N_A entities (atoms, molecules,...). A mol of a species of atomic/molecular mass X amu contains has a mass equal to X grams. The number of moles in a system is equal to the number of atoms $\mathcal{N}$ divided by N_A : $\mathcal{N}/N_A$ [5]

1.4. Gases

Equations of state

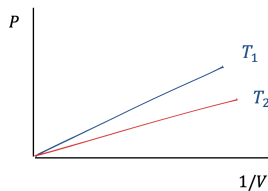
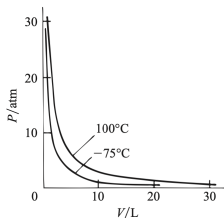
Equations of state are relationships between state variables.

Robert Boyle, in 1662, established a first relationship between state variables for a gas, i.e. an (incomplete) equation of state [6]

Boyle's law

$$PV = f(T, N)$$

where f is a function of temperature and amount of matter.



1.4. Gases

Behavior of ideal gases

Boyle's Law

Boyle's law tells us that if T and N are constant, PV is constant.

$$P_1 V_1 = P_2 V_2$$

Laws of Charles and of Gay-Lussac

Jacques Charles[7] made other experiments with gases at constant P and N , varying the temperature, around 1780, which were verified by Joseph Louis Gay-Lussac and John Dalton, some years later. According to this,

$$\frac{V}{T} = g(P, N)$$

Namely, if P i N are constant, then

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

1.4. Gases

Avogadro's hypothesis and thermal equation of state

It was not until 1811, inspired by the incipient Kinetic Theory of Gases, that Amedeo Avogadro formulated his hypothesis: [4]

The same volume of different gases at the same pressure contain the same number of particles.

We can gather the results of Boyle, Charles and Avogadro into a single equation,

$$PV = NRT \quad \text{Thermal equation of state} \quad (1)$$

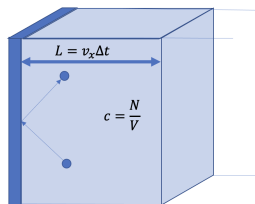
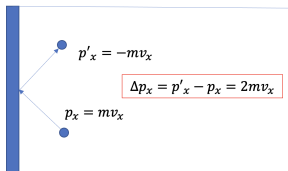
where N is the number of moles of molecules, and T is here an absolute temperature (this can be seen from the fact that $T = 0$ implies $P = 0$). Constant R is called the **gas constant**:

- ▶ $R = 8,3144621(75) \text{ J/K mol}$
- ▶ $R = 1.987 \text{ cal /K mol}$
- ▶ $R = 0.082 \text{ atm l/K mol}$

1.4. Gases

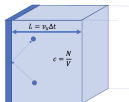
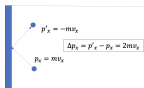
Kinetic theory and pressure

Like temperature, pressure also has a microscopic interpretation: the force exerted by the collisions of the gas onto the walls.



1.4. Gases

Kinetic theory and pressure



If we take an arbitrary Δt , we can calculate how many collisions will occur and compute the pressure easily. The total change in momentum in this interval will be Δp_x times *half* of all particles inside the blue cube on the right,

$$\Delta p_x^{\text{tot}} = \frac{1}{2} \frac{\mathcal{N}}{V} A L \Delta p_x$$

From Newton's second law, if Δt is very small, then $F = \Delta p_{\text{tot}} / \Delta t$. If the pressure is $P = F/A$, then we will have,

$$P = \frac{F}{A} = \frac{\frac{1}{2} \frac{\mathcal{N}}{V} A (v_x \Delta t) 2m v_x}{A \Delta t} = \frac{\mathcal{N}}{V} m \langle v_x^2 \rangle = \frac{\mathcal{N}}{V} k_B T = \frac{N}{V} R T \quad \text{q.e.d.}$$

1.4. Gases

Pressure units

Throughout the course we will preferably use the International System of Units (or SI from French 'Système international d'unités'), based on the basic units of mass (kg), length (m), time (s), charge (C), temperature (K) . . . , from which the others are derived, such as those of force ($N = \text{kg m/s}^2$), energy ($J = \text{kg m}^2/\text{s}^2$), etc.

Pressure is defined as a force per unit area. If we take units of the SI, the unit of pressure is N/m^2 which has been named **Pascal (Pa)**.

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

The *bar* is a multiple, not a ISU unit itself.

$$1 \text{ bar} = 10^5 \text{ Pa}$$

We use it because atmospheric pressure is approximately 1 bar.

Other frequently used units are

- ▶ $1 \text{ atm} = 101325 \text{ Pa} \simeq 1 \text{ bar}$
- ▶ $1 \text{ kgf/cm}^2 = 9.8 \times 10^4 \text{ Pa} \simeq 1 \text{ bar}$
- ▶ $1 \text{ psia} = 1 \text{ lbf/in}^2 = 6895 \text{ Pa}$ (*psia: pounds force absolute per square inch*)

1.4. Gases

Volume units

Volume measurements have been made since ancient times and, for this reason, there are many and very varied ones, with local variations.

The SI measure of volume is the m^3 .

Other common measures of volume are

- ▶ $1\text{ cm}^3 \times \left(\frac{1\text{ m}}{100\text{ cm}}\right)^3 = 10^{-6}\text{ m}^3$
- ▶ $1\text{ l} = 10^{-3}\text{ m}^3$
- ▶ $1\text{ ml} = 10^{-3}\text{ l} = 1\text{ cm}^3$

1.4. Gases

Partial pressure

From a modern point of view, **an ideal gas is a gas of point particles which do not interact with each other and only have kinetic energy (translational, rotational, vibrational, etc.).**

If we have a mixture of gases under these conditions, we expect the total pressure to depend only on the total number of particles present, regardless of their nature. Then, if we have a mixture of $i = 1, \dots, n$ gases, with N_i moles of the i -th component, the total number of particles (moles of particles) will be

$$N = \sum_{i=1}^n N_i$$

Boyle's law will then be written as

$$P = \frac{NRT}{V} = \left(\sum_{i=1}^n N_i \right) \frac{RT}{V} = \left(\sum_{i=1}^n \frac{N_i RT}{V} \right) = \sum_{i=1}^n P_i$$

1.4. Gases

Partial pressure

The quantities P_i are called **partial pressure of component i** and refer to the pressure that the i -th component would exert if it were alone.

We can also express the partial pressure in another way, by introducing the mole fractions, defined as

$$x_i = \frac{N_i}{N},$$

és a dir,

$$P_i = \frac{N_i RT}{V} = \frac{N_i}{N} \frac{NRT}{V} = x_i P$$

These expressions are useful.

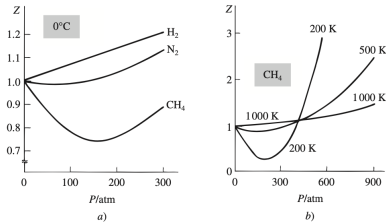
1.4. Gases

Compressibility factor and deviation from ideality

The compressibility factor

$$Z(T, P) \equiv \frac{P V(T, P)}{NRT}$$

has been used to identify the deviation of real gases from ideality, because for ideal gases its value is a constant, $Z = 1$, regardless of the pressure.



1.4. Gases

Compressibility factor and deviation from ideality

Real gases behave approximately as ideal under conditions of

- ▶ low pressure (low density implies little interaction), and
- ▶ high temperature (predominance of kinetic over potential energy);

with the ideal gas model being suitable in many everyday situations. However, for example, it does not predict the condensation of gases into liquids (which involves the attraction between the atoms, molecules).

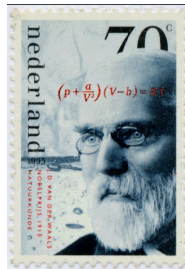
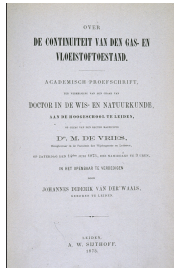
1.4. Gases

Compressibility factor and deviation from ideality

An equation of state that describes the behavior of real gases is the van der Waals equation

$$P = \frac{NRT}{V - Nb} - a \left(\frac{N}{V} \right)^2$$

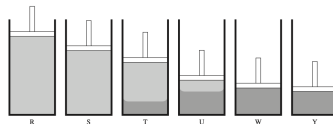
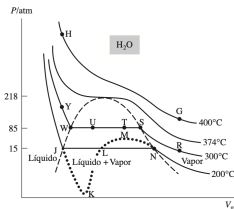
This is the first *cubic* equation of state, which depends on two parameters, a and b . van der Waals thought that parameter b was the **molecular volume**, while parameter a captured the attractive interaction between the molecules (nowadays called **van der Waals forces**) (Doctoral Thesis of Johannes D. van der Waals, 1874 Leiden).



1.4. Gases

Law of corresponding states

The unique feature of the van der Waals equation is in the title of his doctoral thesis *On the continuity between the gaseous and liquid state*, because it allowed him to describe the phase transition between liquid and gas:



1.4. Gases

Law of corresponding states

van der Waals observed that the behavior predicted by his equation of state allowed all gases to be assimilated to a single behavior. (Though later this was seen not to be the case, his equation works for a good number of substances.) It is what is called **the law of the corresponding states**. This can be seen by introducing variables which are called *reduced*,

$$v^* = \frac{V}{3Nb} = \frac{V}{V_c}; \quad P^* = \frac{27b^2}{a} P = \frac{P}{P_c}; \quad T^* = \frac{27b}{8a} RT = \frac{T}{T_c}$$

The parameters V_c , P_c and T_c define the critical point of the model.

With these new variables, the vdW equation of state remains independent of the parameters (a and b): *master curve*.

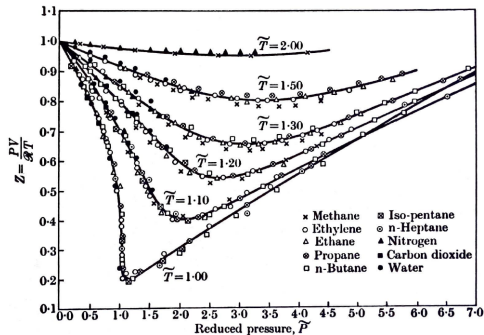
$$P^* = \frac{\frac{8}{3} T^*}{v^* - \frac{1}{3}} - \frac{3}{v^{*2}}$$

This is the key point of the law of the corresponding states: **All gases which depend on two parameters, behave in the same way when we express them in terms of the reduced variables** by using the values of the critical point.

1.4. Gases

Compressibility factor and deviation from ideality

We find that even gases which do not obey the vdW equation may behave similarly when their P , V , and T are expressed as reduced variables (nondimensionalized with the critical point values). As shown below, the Z vs P^* data collapse onto a single curve which depends on the reduced temperature T^* , for gases which are apolar or do not present significant asymmetries.



1.4. Gases

Virial expansion

The van der Waals equation was the first example of a *cubic* equation of state, and became the basis for many other equations (very commonly used in ChemE) such as Peng-Robinson, Redlich-Kwong, etc.

There is, though, a general expression for small deviations from ideality (the ideal gas works at high T and $1/v \rightarrow 0$ where $v \equiv V/N$), which we can characterize as a Taylor expansion in powers of $1/v$ which we call **virial**:

$$P(v, T) = \frac{RT}{v} \left(1 + A_2(T) \frac{1}{v} + A_3(T) \frac{1}{v^2} + \dots \right)$$

The A_i are called the **virial coefficients**.

The temperature at which $A_2(T_B) = 0$ (second virial coefficient) is called the Boyle temperature, at and around which the system behaves almost ideally up to remarkably high densities.

Recommended reading

- [1] I. Levine, *Principios de Fisicoquímica*, McGraw-Hill; Chapter 1; real gases: beginning of Chap. 8
- [2] H.B. Callen, *Termodinámica*, Ediciones AC
- [3] https://en.wikipedia.org/wiki/Daniel_Gabriel_Fahrenheit
- [4] https://ca.wikipedia.org/wiki/Nombre_d%27Avogadro
- [5] [https://en.wikipedia.org/wiki/Mole_\(unit\)](https://en.wikipedia.org/wiki/Mole_(unit))
- [6] https://ca.wikipedia.org/wiki/Llei_de_Boyle
- [7] https://ca.wikipedia.org/wiki/Llei_de_Charles_i_Gay-Lussac